MIL-S-24154A(SHIPS)
10 March 1967
SUPERSEDING
MIL-S-24154(SHIPS)
30 July 1965
(See 6.8)

MILITARY SPECIFICATION

SYNTACTIC BUOYANCY MATERIAL FOR

HIGH HYDROSTATIC PRESSURES

1. SCOPE

1.1 Scope. - This specification covers requirements for rigid syntactic buoyancy materials used to provide buoyancy for deep ocean submerged research vehicles, and other related applications.

1.2 Classification. - The syntactic byoyancy material shall be of the following types, as specified (see 6.2).

Type I - For 4,500 p.s.i.g. hydrostatic pressure.
Type II - For 10,000 p.s.i.g. hydrostatic pressure.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of the specification to the extent specified herein.

STANDARD

MILITARY
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables For Percent Defective.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9330

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D638 - Tensile Properties of Plastics.
D695 - Compressive Properties of Rigid Plastics.
D732 - Shear Strength of Plastics.
D1622 - Apparent Density of Rigid Cellular Plastics.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N. Y. 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. - The syntactic buoyancy material furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.6).

3.1.1 For qualification, the physical properties of the syntactic buoyancy material shall conform to Tables I and II before and after exposure to the test cycles of 4.6.6 and 4.6.7.1.

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Table I - Physical requirements before and after 1,000 cycle hydrostatic exposure (qualification tests only).

Property[1/]	Type I		Type II	
	Before exposure	After exposure[2/]	Before exposure	After exposure[3/]
Density (max.) (block) (lbs/cu.ft.) module with skin	38	--	44	--
Density (max.) (lbs/cu.ft.) specimen	38	--	44	--
Water absorption (max.) (percent by weight)	--	3	--	3
Uniaxial ultimate compressive strength lbs/sq.in. (min.)	7,000	6,000	12,500	11,500
0.2 percent offset uniaxial compressive stress, lbs./sq.in. (min.)	5,000	4,250	10,000	9,000
Compressive modulus lbs./sq.in. (min.)	300,000	250,000	425,000	400,000
Tensile strength lbs./sq.in. (min.)	1,500	--	4,000	--
Shear strength lbs./sq.in. (min.)	2,000	--	4,000	--
Bulk modulus (min.) lbs./sq.in.	275,000	--	350,000	--
Hydrostatic compressive strength, lbs./sq.in. (min.)	9,000	--	16,500	--
Hydrostatic compressive strength, max. reduction from initial before exposure strength, percent	--	10	--	10

1/ Density, initial uniaxial compressive strength, initial 0.2 percent offset uniaxial compressive stress, and initial compressive modulus can be determined from one set of specimens; also, water absorption, after exposure uniaxial compressive strength, after exposure 0.2 percent offset uniaxial compressive stress, and after exposure compressive modulus can be determined from another set of specimens.

2/ After 1,000 cycles of hydrostatic exposure between 0 to 4,500 p.s.i.g.

3/ After 1,000 cycles of hydrostatic exposure between 0 to 10,000 p.s.i.g.

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3.2 Physical properties. - The physical properties of the syntactic buoyancy material shall conform to the requirements specified in table II.

Table II - Physical requirements before and after one week of static exposure (qualification and quality conformance tests.

Property[1]	Type I		Type II	
	Before exposure	After exposure[2]	Before exposure	After exposure[3]
Density (max.) (lbs/cu.ft.) module with skin	38	--	44	--
Density (max.) (lbs/cu.ft.) specimen	38	--	44	--
Water absorption (max.) (percent by weight)	--	3	--	3
Uniaxial ultimate compressive strength lbs/sq.in. (min.)	7,000	6,300	12,500	12,000
0.2 percent offset uniaxial compressive stress, lbs./sq.in. (min.)	5,000	4,500	10,000	9,500
Compressive modulus lbs./sq.in. (min.)	300,000	270,000	425,000	400,000
Tensile strength lbs./sq.in. (min.)	1,500	--	4,000	--
Shear strength lbs./sq.in. (min.)	2,000	--	4,000	--
Bulk modulus (min.) lbs./sq.in.	275,000	--	350,000	--
Hydrostatic compressive strength, lbs./sq.in. (min.)	9,000	--	16,500	--
Hydrostatic compressive strength, max. reduction from initial before exposure strength, percent	--	10	--	10

1/ Density, initial uniaxial compressive strength, initial 0.2 percent offset uniaxial compressive stress, and initial compressive modulus can be determined from one set of specimens; also, water absorption, after exposure uniaxial compressive strength, after exposure 0.2 percent offset uniaxial compressive stress, and after exposure compressive modulus can be determined from another set of specimens.

2/ One week (168 hours) exposure at 4,500 p.s.i.g. hydrostatic pressure.

3/ One week (168 hours) exposure at 10,000 p.s.i.g. hydrostatic pressure.

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3.3 Material. - The syntactic buoyancy material (see 6.3) shall consist of a low density filler, such as hollow glass microspheres, incorporated in a resin matrix, such as epoxy resin.

3.4 Sizes and forms. - The syntactic buoyancy material shall be furnished in modules, blocks, sheet, or other forms, as specified (see 6.2).

3.5 Dimensional tolerances. - The dimensions and dimensional tolerances shall be as specified (see 6.2).

3.6 Color. - The color shall be the natural color of the product.

3.7 Resin skin. - Syntactic buoyancy material shall be supplied in the shapes and sizes required with or without resin skin or with a specific number of sides with or without resin skin, as specified (see 6.2). The resin skin normally results from the production process. In the event that syntactic buoyance material with resin skin is specified, and the natural skin is damaged, it shall be replaced using the same resin system as the cured foam casting, or a resin system approved by the command or agency concerned.

3.8 Workmanship. - The syntactic buoyancy material shall be uniform in quality and be free of cracks, splits, seams, porous or resin-rich areas.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests.[1/] Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the tests specified in 4.6, and shall meet all the requirements of tables I and II.

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.6 and 6.7).

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4.2.1 Qualification sample. - A one-piece casting (see figure 1) of syntactic buoyancy material, measuring 12 by 12 by 12 inches plus or minus 1/4 inch with a resin skin shall be submitted for qualification inspection. The qualification sample shall be a single homogenous block and shall not be composed of smaller modules adhered together.

4.3 Qualification inspection. - Unless otherwise specified (see 6.2), qualification inspection shall consist of the examination specified in 4.5.2.

4.4 Sampling. -

4.4.1 Sampling for qualification inspection. - Nine each, ilitial and exposure, compressive, water absorption, shear, and density specimens shall be selected from the 12 by 12 by 12 inch qualification sample. Two 1 inch diameter by 2 inch long specimens shall be randomly selected from each of the following nine stratified sections: top-center, top-edge, top-corner, middle-center, middle-edge, middle-corner, bottom-center, bottom-edge, and bottom-corner; figure 1 graphically shows stratification. This provides nine specimens for determining density and initial compressive properties and nine for providing water absorption and after exposure compressive properties, representing each of the non stratified sections. Five shear specimens shall be randomly selected from the top-corner specimen, middle-corner, bottom-corner, middle-edge, and middle center. Five tensile, three bulk modulus and three hydrostatic compressive strength specimens shall be randomly selected from the remaining 12 by 12 by 4 inch section shown on figure 1.

4.4.2 Sampling for quality conformance inspection. - All modules shall be inspected for conformance with 4.5.2.

4.4.3 Sampling for quality conformance testing. - From each lot, modules shall be selected in accordance with MIL-STD-414, Section B at an A.Q.L. of 4.0 percent. Modules tested shall represent largest (in volume) production sizes, unless otherwise specified (see 6.2).

4.4.4 Selection of individual specimens. - From each sample module selected in 4.4.3, one specimen shall be taken for each of the destructive tests specified 4.5.3 (see footnote 1/ of table II for properties which can be determined with an individual specimen). The total number of specimens for each test shall contain an approximately equal number representing the top, middle, and bottom of the sample modules. For example, specimens shall be selected from the top of one module, the middle of the second module, the bottom of the third module, etc., to obtain a representative sample.

4.4.5 Acceptable quality level. - Lots shall be accepted or rejected on the basis of an acceptable quality level (A.Q.L.) of 4.0 percent for each test separately. Procedures of MIL-STD-414, Section B shall be followed.

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4.5 Quality conformance inspection and tests. -

4.5.1 Lot. - For the purpose of sampling, examination and test, a lot shall consist of one batch of microspheres and resin mixed at the same time in the same mixer and subsequently poured into molds to form one or more modules.

4.5.2 Examination. - Each module shall be examined for size and form, dimensional tolerances, and defects and workmanship to determine conformance with the requirements of this specification. Any module found not to be in conformance with the requirements of this specification shall be cause for rejection.

4.5.3 Quality conformance tests. - Quality conformance tests shall consist of the tests specified in 4.6 except that 4.6.5.1 shall not apply. Physical properties of test specimens shall meet all the requirements of 3.3 (table II). The tests of 4.6.9 and 4.6.10 shall be run only on the first, middle, and last lot or as specified (see 6.2).

4.6 Test methods. - Each individual test value shall be reported.

4.6.1 Selection of test specimens. - Test specimens shall be selected as specified in 4.7.1 for qualification tests and 4.7.2 for quality conformance tests.

4.6.2 Standard test specimen. - Unless otherwise specified (see 6.2), the standard 1-inch diameter by 2-inch long cylindrical test specimen (1" ± 0.005" dia. x 2" ± 0.005" long) shall be used. The dimensional tolerance shall be plus or minus 0.1 inch.

4.6.3 Conditioning of test specimens. - Test specimens shall be conditioned in a standard laboratory atmosphere of 23° ± 1.1°C. (73.4° ± 2°F.) and 50 ± 5 percent relative humidity for a minimum of 24 hours prior to testing.

4.6.4 Test conditions. - Tests shall be conducted in a standard laboratory atmosphere of 23° ± 1.1°C. (73.4° ± 2°F.) and 50 ± 5 percent relative humidity unless otherwise specified in the testing method or in this specification.

4.6.5 Tests for density. -

4.6.5.1 With resin skin. - Prior to sectioning the sample, the overall density of the sample with the skin (see 4.2.1) shall be determined. Measure the sample to the nearest 0.1 inch and calculate the volume in cubic inches, and weigh to the nearest 1.0 gram. Calculate density by:

$$\frac{\text{Weight of sample (grams)}}{\text{Volume of sample (cu. in.)}} \times 3.81 = \text{Density (lbs./cu.ft.)}$$

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4.6.5.2 Density of specimens (without resin skin). - One-inch diameter by 2-inch long cylindrical test specimens shall be used to determine the density of the syntactic material in accordance with ASTM D1622.

4.6.6 Exposure tests. - Before physical tests are performed, the appropriate specimens shall be exposed continuously in a liquid for one week at 4,500 p.s.i.g. hydrostatic pressure for type I and at 10,000 p.s.i.g. hydrostatic pressure for type II. Unless otherwise specified (see 6.2), the exposure liquid for immersion tests shall be tap water.

4.6.7 Liquid absorption. - Expose 1-inch diameter by 2-inch long test specimens to test liquid at 75° ± 10°F. for 10 seconds (see 4.6.6). Wipe specimen dry in a paper towel and immediately weigh to the nearest milligram. Expose specimens for one week in the test liquid at 4,500 p.s.i.g. hydrostatic pressure for type I and at 10,000 p.s.i.g. hydrostatic pressure for type II. Remove specimens from the test fluid, wipe dry in a paper towel and immediately weigh to the nearest milligram. Calculate liquid absorption by:

$$\frac{\text{Final weight - initial weight}}{\text{Initial weight}} \times 100 = \text{percent liquid absorption by weight}$$

4.6.7.1 1,000-cycle fatigue test (for qualification only).- Percent water absorption after exposure to a 1,000-cycle fatigue test shall be obtained after exposure to the following fatigue cycle:

(a) Expose 1-inch diameter by 2-inches long test specimens to test liquid at 75° ± 10°F. for 10 seconds. Wipe specimen dry in a paper towel and immediately weigh to the nearest milligram. Expose specimens for 1,000 cycles in the test liquid in accordance with the following test cycle:
   1. 4 minute rise in hydrostatic pressure:
      From atmospheric to 4,500 p.s.i.g. for type I only.
      From atmospheric to 10,000 p.s.i.g. for type II only.
   2. Hold pressure at:
      4,500 p.s.i.g. for 12 minutes for type I only.
      10,000 p.s.i.g. for 12 minutes for type II only.
   3. Release pressure at a rate of approximately 10 seconds/1000 psi so that:
      Pressure is reduced to atmospheric in 45 ± 10 seconds for type I material only.
      Pressure is reduced to atmospheric in 100 ± 20 seconds for type II material only.
   4. Hold 4 minutes at atmospheric pressure.
   5. Repeat for 1,000 cycles.

(b) Remove specimens from the test fluid after completion of the 1,000 cycle tests. Wipe dry in a paper towel and immediately weigh to the nearest milligram. Calculate water absorption in accordance with formula in 4.6.7.

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4.6.8 Uniaxial compressive tests.- One-inch diameter by 2-inch long test specimens shall be tested in accordance with ASTM D695 to determine the compressive strength, 0.2 percent offset compressive stress, and the comoressive modulus.

4.6.9 Tensile strength.- The tensile strength test shall be in accordance with ASTM D638 using one-half inch thick, type II specimens as described therein. (See 6.4).

4.6.10 Shear strength.- The shear strength shall be determined in accordance with ASTM D732.

4.6.11 Bulk modulus.- This method describes the procedure for determining the bulk modulus of a syntactic buoyancy material by means of a piston volume displacement method.

4.6.11.1 Apparatus.- The following apparatus shall be used.

Cylinder and piston (see 6.5).
Hydrostatic pressing chamber.
I.D. = $2\frac{1}{4}$ inches.
O.D. = $4\frac{1}{2}$ inches.
Working height = 10 inches.
Pressure rating = 15,000 p.s.i.g.
Tonnage required = 29.8T.
Chamber material - AISI 4340 H.T. to 140,000 p.s.i., Y.S.
Piston seal: Buna N "O" ring with teflon backups.

4.6.11.2 Test specimen.- The size of the test specimen shall be 4 by $1\frac{1}{2}$ by $1\frac{1}{2}$ inches (148 ml.).

4.6.11.3 Procedure.- The procedure shall be as follows:

(a) The high grade alloy steel cylinder is partially filled with 500 ml. of water (fresh distilled water), and a stainless steel block measuring 4 by 1-1/2 by 1-1/2 inches (148 ml) is inserted into the chamber.

(b) The water in the cylinder is then compressed by the piston to 4,500 p.s.i.g. on a compression machine for type I and to 10,000 p.s.i.g. for type II using a rate of cross-head travel of 0.05 in./min.

(c) The piston displacement and hence the change in volume of the water is recorded automatically on the machine chart. (A dial gage may be used in lieu of a recorder).

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(d) The steel block is removed and the syntactic specimen is introduced into the cylinder with the same volume of water present.

(e) The combination of water and syntactic specimen is also subjected to 4,500 p.s.i.g. hydrostatic pressure on the compression machine for type I and to 10,000 p.s.i.g. for type II using a rate of cross-head travel of 0.05 in./min.

4.6.11.4 Calculations.- Calculations shall be made as follows:

(a) The volume change of the syntactic specimen is then determined by substracting the compression of the water from the compression of the combination.

(b) A correction is made for water absorption by weighing the specimen before and after exposure. (The weight gain is converted to volume and substracted from the actual ΔV_s below.)

(c) The relationship between the volume change of water and volume change of the syntactic material may be expressed in the following way:

$$k = -\frac{\Delta V_s}{V_s} = -\left(\frac{\Delta V_c - \Delta V_w}{V_s}\right)$$

Where: k = compression.
ΔP = change in pressure
ΔV_s = change in volume of syntactic material (corrected for absorbed water).
ΔV_c = change in volume of combination, syntactic material and water.
ΔV_w = change in volume of water.
V_s = original volume of syntactic material.

Where: Bulk modulus is $E_B = \frac{-\Delta P}{\Delta V_s} - V_s$

4.6.12 Hydrostatic compressive strength, - The hydrostatic compressive strength test shall be determined as specified in Appendix A.

4.7 Inspection for preparation for delivery. - Packaging, packing, and marking shall be inspected to assure compliance with section 5 of this specification.

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early use. -

5.1.1 Syntactic buoyancy material.-

5.1.1.1 Packaging.- Packaging, which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier, at lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

6. NOTES

6.1 Intended use.- Syntactic buoyancy material is intended for use in submerged research vehicles, underwater floats, encapsulants, etc. As such, it will be directly exposed to the sea or some other liquid medium which provides hydrostatic forces on the exposed surfaces.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Size, form, and number of modules required (see 3.5).
(d) Dimensions and dimensional tolerances required (see 3.6).
(e) Requirements for resin skin (see 3.8).
(f) If qualification inspection is other than specified (see 4.3).
(g) If representation of modules is other than specified (see 4.4.3).
(h) Frequency of tensile and shear tests, if other than specified (see 4.5.3).
(i) If standard test specimen is other than specified (see 4.6.2).
(j) When exposure liquid other than tap water is required (see 4.6.6 and 4.6.7).
(k) Packaging, packing and marking if other than specified (see 5.1).

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6.3 Definition.- Syntactic buoyancy material (or syntactic foam, as known in The Trade) is defined as a material consisting of hollow sphere fillers incorporated in a resin matrix.

6.4 Type I specimens as described in ASTM D638 have been found unsuitable for tensile strength tests because syntactic material is brittle, and type I specimens have occasionally broken in the jaws.

6.5 Apparatus may be procured from Loomis Engineering Company, Caldwell, New Jersey, or equal.

6.6 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 24154 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Naval Ship Engineering Center, Department of the Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.7).

6.7 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.8 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

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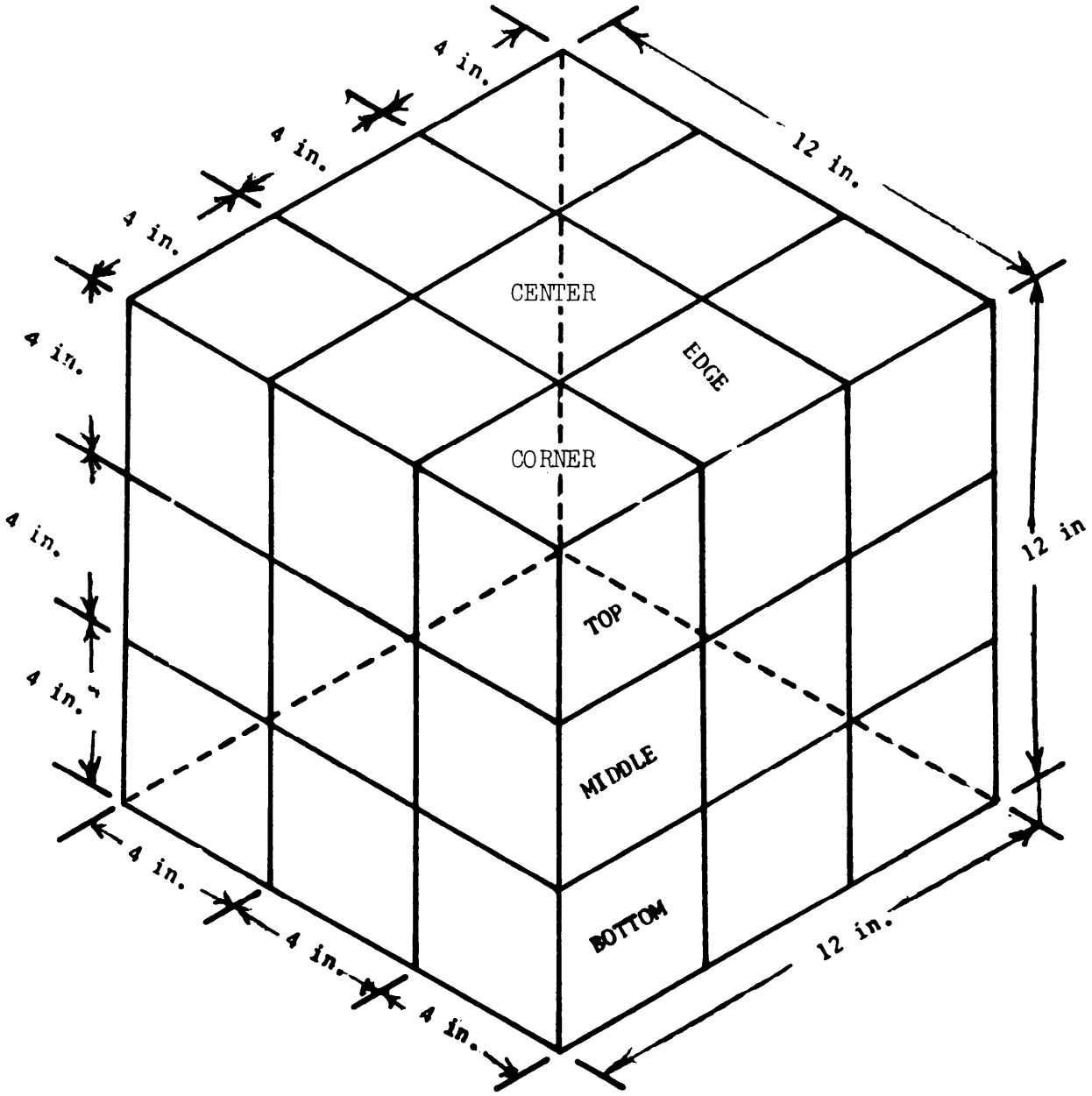


Figure 1 - Stratification of qualification sample.

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APPENDIX A

10. SCOPE

10.1 This test procedure is designed to quickly and accurately determine the hydrostatic compressive strength of any syntactic foam capable of withstanding 1000 psi or higher hydrostatic loads. The test is simple, yet accurate to within 100 psi.

20. SUMMARY OF METHOD

20.1 This test method consists of applying a hydrostatic compressive load to a syntactic foam specimen via a test machine cross-head which bears down on a sliding piston contained in a high pressure cylinder. The load required to cause failure of the specimen shall be taken from a load-deflection plot or the applied load as observed on the test machine dial.

30. SIGNIFICANCE

30.1 This test provides information about the usefulness of a syntactic foam for a given hydrostatic pressure application. The test is limited to foams with a hydrostatic compressive strength of more than 1000 psi because the accuracy of determinations below this level are questionable.

40. INTERFERENCES

40.1 Some of the problems which can be encountered in this method and which should be avoided are:

(a) Entrapped air in the test vessel.
(b) Leaks in the hydraulic system.
(c) Friction between the seal and the cylinder walls.

40.2 The test procedure is to simply place the test specimen of standard size into the test vessel of standard size containing a standard volume of water, bleed off the entrapped air and apply a load at a standardized rate of speed (cross-head speed) with a test machine and plot a load-deflection curve.

50. APPARATUS

50.1 Any suitable testing machine capable of control of constant rate-of-crosshead movement and consisting of a drive mechanism, a load indicator and an extensometer shall be used. A test machine such as specified in method ASTM D695, Test for Compressive Properties, can be used.

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50.2 Piston-cylinder apparatus. - A test vessel consisting of a piston-cylinder apparatus, size 1.5 inches inside diameter by 6 inches deep (working dimensions) such as illustrated in figure 1 may be used. The apparatus shall be rated for 35,000 psi minimum. (NOTE: A 1.5 inch by 6 inches deep (working-dimensions) test vessel rated for 35,000 psi supplied by Loomis Engineering and Manufacturing Co., Caldwell, N. J., or equivalent, shall be used).

60. SAFETY PRECAUTIONS

60.1 The usual safety precautions applicable to systems under high hydraulic loads shall be observed.

70. SAMPLE SIZE

70.1 The sample shall consist of a 1 inch ± 0.1 inch diameter cylinder, 2 inches ± 0.1 inch long.

80. PROCEDURE

80.1 The hydrostatic compressive specimen shall be placed in the isostatic pressing cylinder partially filled with 210 ml. of water.

80.2 The piston shall be placed in the cylinder and the entrapped air and 10 ml. of water shall be bled off from the vent hole, and the air vent shall then be tightly closed.

80.3 The piston cylinder shall be loaded on a compression machine using a rate of cross-head travel of 0.20 in./min. (Preliminary investigations indicate that hydrostatic compressive strength is not significantly affected by load rate.)

80.4 The hydrostatic compressive load shall be that point at which a substantial drop in load is evidenced on the compression machine's dial gage or the load-deflection curve.

80.5 The load shall be released upon reaching the hydrostatic compression load and the equipment shall be prepared for the next test.

90. INTERPRETATION OF RESULTS:

90.1 In the case of high strength foams, the failure point is quite obvious, manifested by a sharp break in the load-deflection curve (or a sudden drop in applied load on the test machine dial). Lower strength foams may fail by gradual compression or water absorption and this mechanism of failure is not immediately obvious. This is where the load-deflection curve shall be employed to determine actual failure by determining what percent offset yield corresponds to the arbitrarily chosen water absorption or specimen deformation corresponding to specimen failure.

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10. CALCULATIONS

10.1 Ultimate hydrostatic crush strength shall be calculated (in psi) from the applied hydrostatic load (indicated by the test machine dial or the arbitrarily chosen yield point of the load-deflection curve) divided by the cross-sectional area of the test vessel's piston:

$$\text{Ultimate hydrostatic strength} = \frac{\text{load (in pounds)}}{\text{cross-sectional area of test vessel piston}}$$

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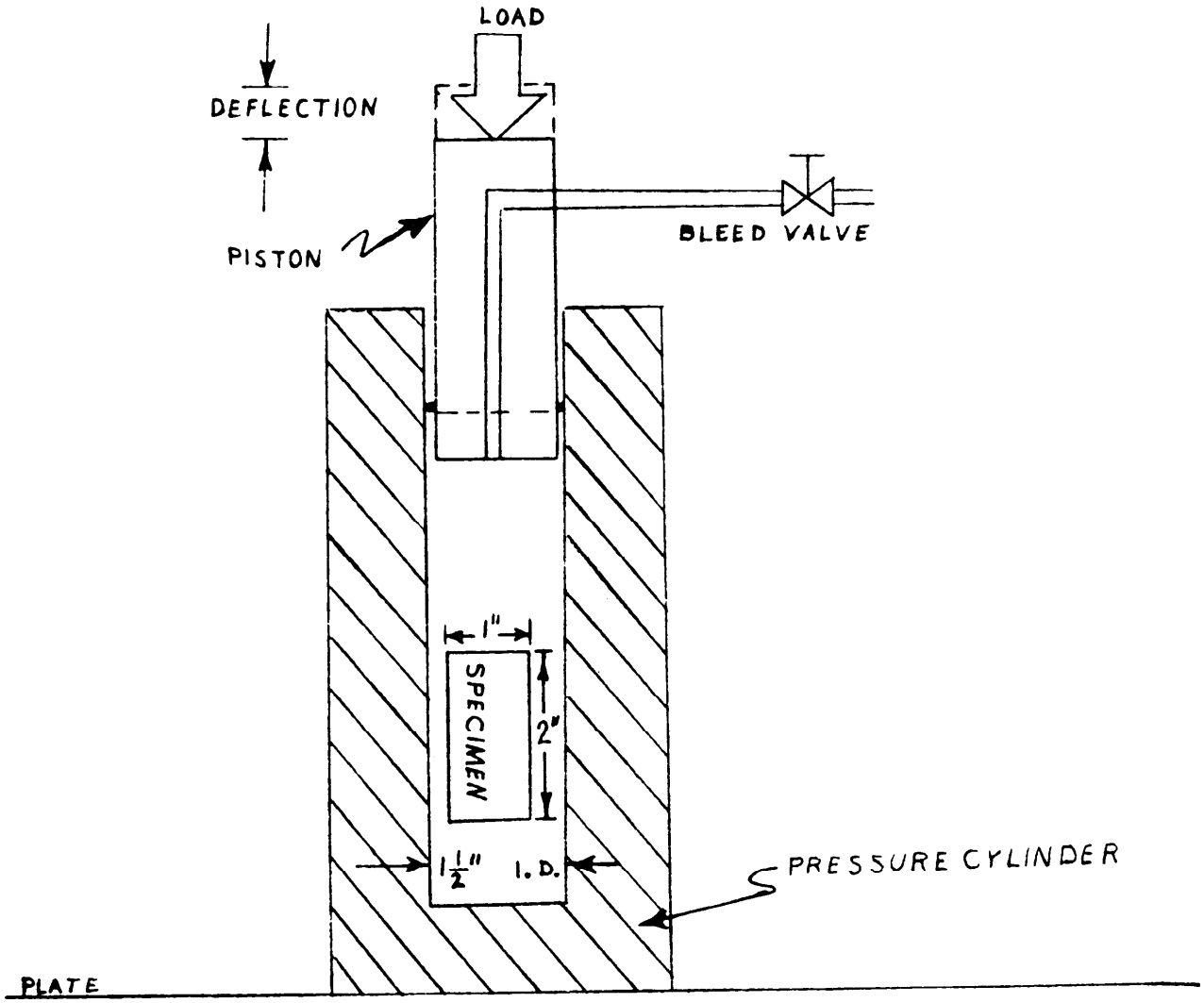


Figure 1A - Schematic of hydrostatic compressive test equipment.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

(NAVSHIPS OVPRT 12-66)

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